

Extern = basically allows variables to be shared globally.

→ extern tells compiler that this variable is defined somewhere.

Proper usage:

config.h

```
extern int max_score;
```

Config.c

```
int max_score = 100;
```

main.c

```
#include "config.h"
```

- Translation units (C-files or .o files that are made at compilation time) need to see where the variables actually are.
- The linker is responsible for combining these files and variables so that global state can be shared across multiple files.
- Also, you can only declare this once. All other files that use that variable need to have 'extern' keyword attached. But sharing global scope is dangerous.

→ Important Note!

- The global variable can be modified outside of the single file scope and that change is reflected everywhere.

Data Types

→ Union

- ↳ Can be used if you need multiple data types in a single variable

↳ `union myData`
`{`

`int myVal;`

`char myName[5];`

`float myFloat;`

`};`

- ↳ Always occupies the memory of the largest
- ↳ Can contain multiple data types but they occupy the same memory space.

→ Size_t

- ↳ Optimization to choose an int that is large enough but not too large.
- ↳ Can't be negative
- ↳ max size depends on the system architecture (64-bit or 32-bit)

Outputs

- Basic output is ofc `printf`

- `fputs` can also be used, has the ability to write to a file.

→ Can be used with fopen and FILE*

fputs(string, FILE*);

Inputs

→ scanf is standard

→ fgets can be used to read from a file. Also can be used to get keyboard input using stdin

fgets (buff, n, stream)

buff = pointer to string buffer where input will be stored

n = maximum number of characters that can be entered.

stream = from where we will be reading inputs.

Operators

↳ sizeof

↳ Extremely useful. Used to size of whatever variable or expression there is

↳ Very useful in finding the number of elements in an array

printf("Number of elements #: %lu", sizeof(arr) / sizeof(arr[0]));

↳ Very much useful in malloc and dynamic programming.

Dot and arrow

↳ The dot is used to reference a member of a class / struct variable / object

struct - var: member;

↳ Arrow is used when we have a pointer of a structure

struct - pointer → member

Comma Operator

↳ It can be used to evaluate multiple expressions from left to right.

int a = (x = 5, y = 10, x + y) will end up with 15.